

Introduction

A sequence diagram is a type of UML (Unified Modeling Language) diagram that shows how objects interact in a given situation. It displays the sequence of messages exchanged between objects and the order in which these interactions occur.



Notations of a Sequence Diagram

Lifeline

A lifeline represents an individual participant in the sequence diagram. It is depicted as a vertical dashed line with a box on top that contains the object's name or class.



Notations of a Sequence Diagram

Actor

An actor represents a user or another system that interacts with the system. It is depicted as a stick figure.



Notations of a Sequence Diagram

Activation Bar

An activation bar (or execution specification) shows the period during which an object is performing an action. It is represented as a thin rectangle on the lifeline.



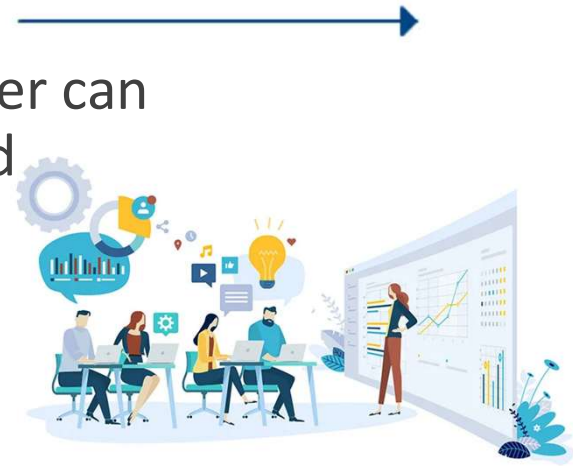
Notations of a Sequence Diagram

Messages

Messages represent communication between objects.

Synchronous Message

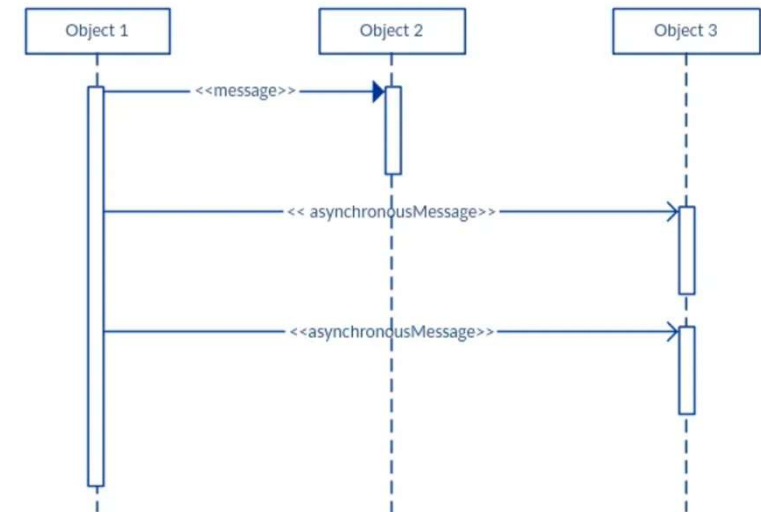
A message that requires a response before the sender can continue. It is represented by a solid line with a filled arrowhead.



Notations of a Sequence Diagram

Asynchronous Messages

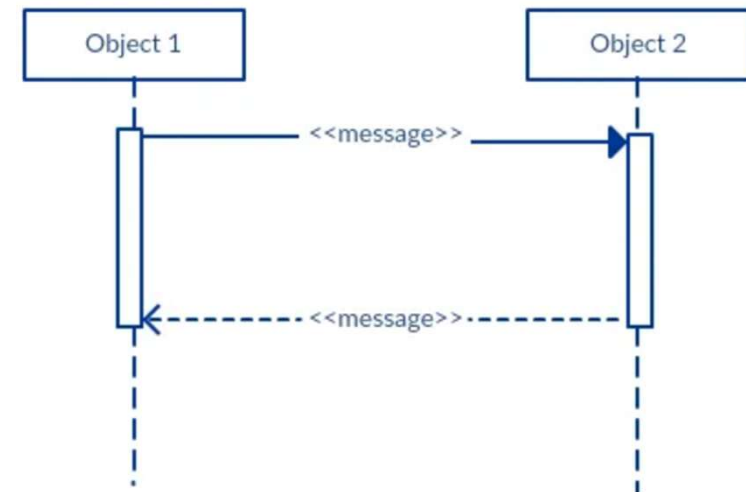
A message that does not require an immediate response. It is represented by a solid line with an open arrowhead.



Notations of a Sequence Diagram

Return Message

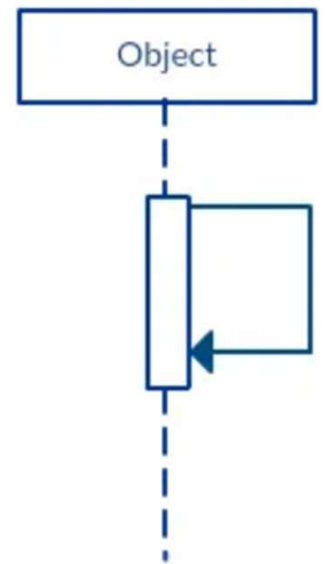
A message that returns a value to the sender. It is represented by a dashed line with an open arrowhead.



Notations of a Sequence Diagram

Self Message/ Reflexive Message

A self-message (or self-call) occurs when an object sends a message to itself. It is represented by a message arrow that loops back to the same lifeline.



Notations of a Sequence Diagram

Combined Fragments

Combined fragments are used to represent conditional logic, loops, or other control structures.

Alt (Alternative)

Represents a choice between different behaviors



Notations of a Sequence Diagram

Opt (Optional)

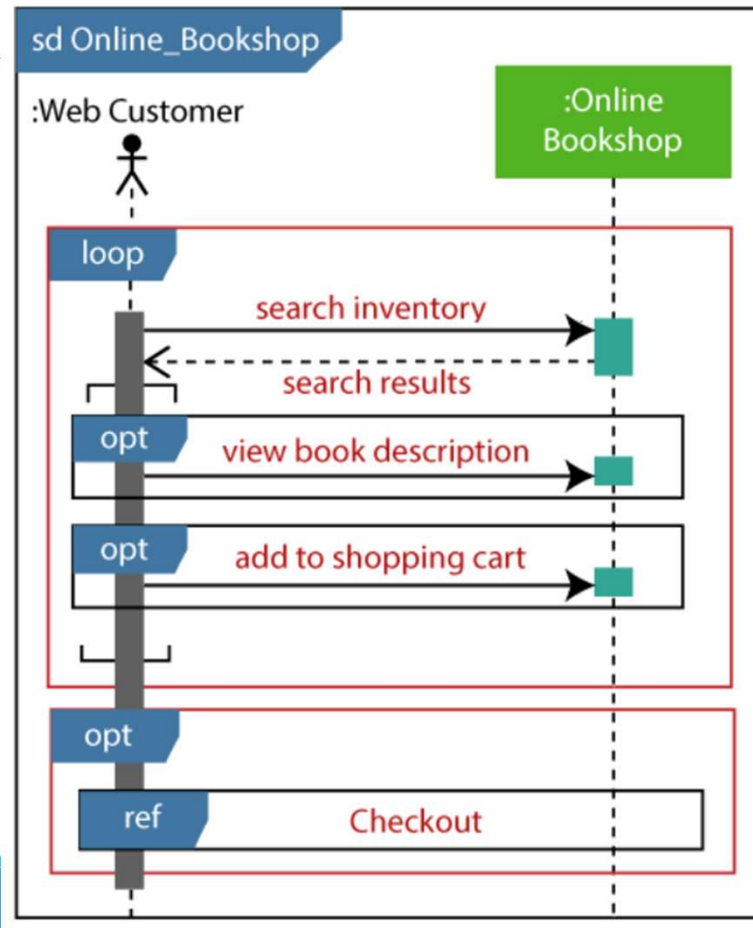
Represents a behavior that will occur only if a certain condition is met.

Loop

Represents a repeated behavior.



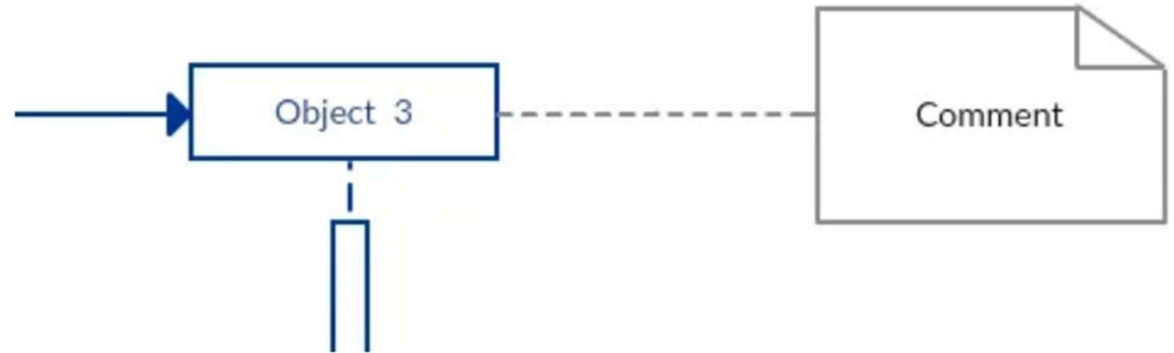
Notations of a Sequence Diagram



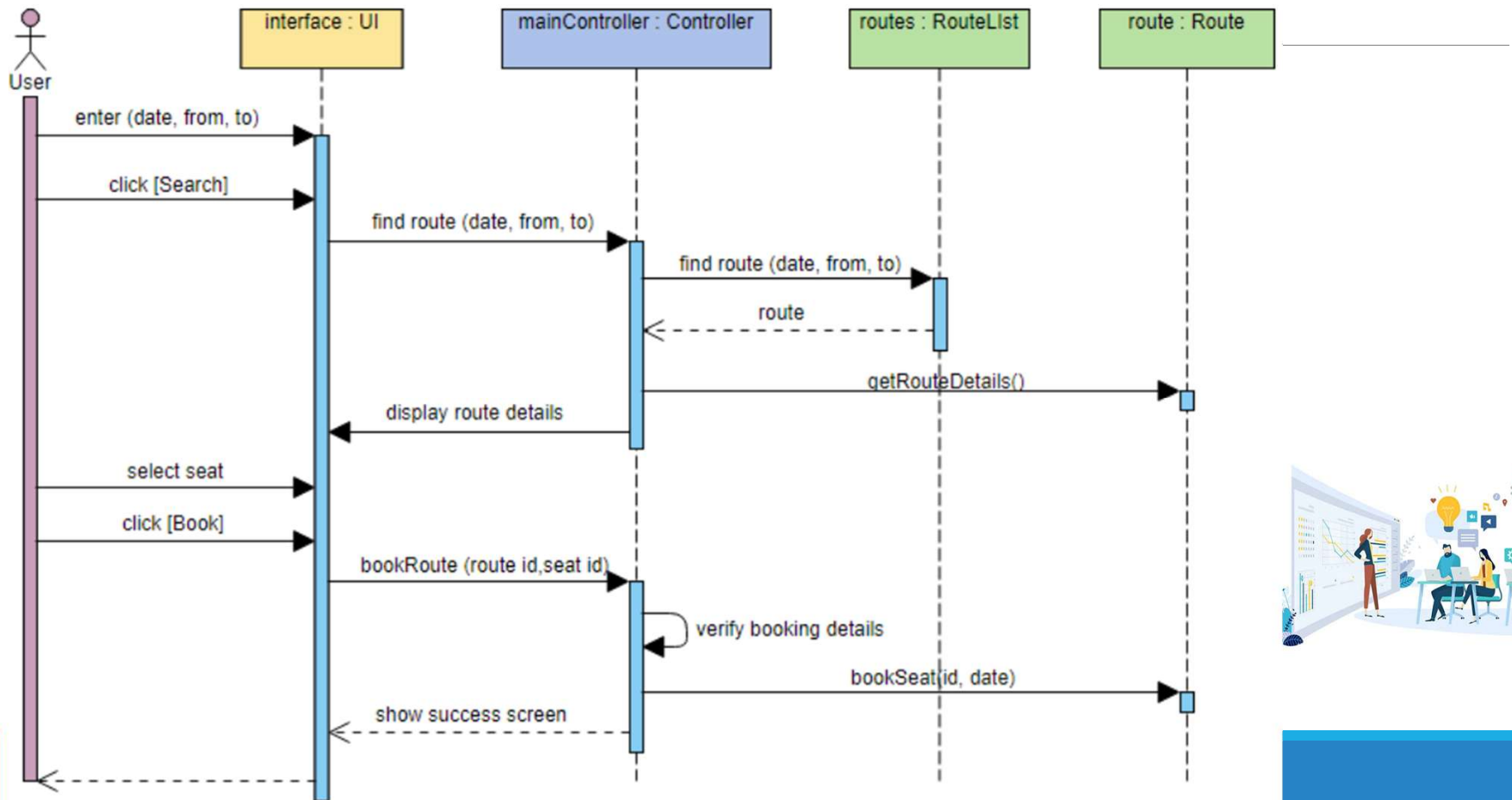
Notations of a Sequence Diagram

Comment

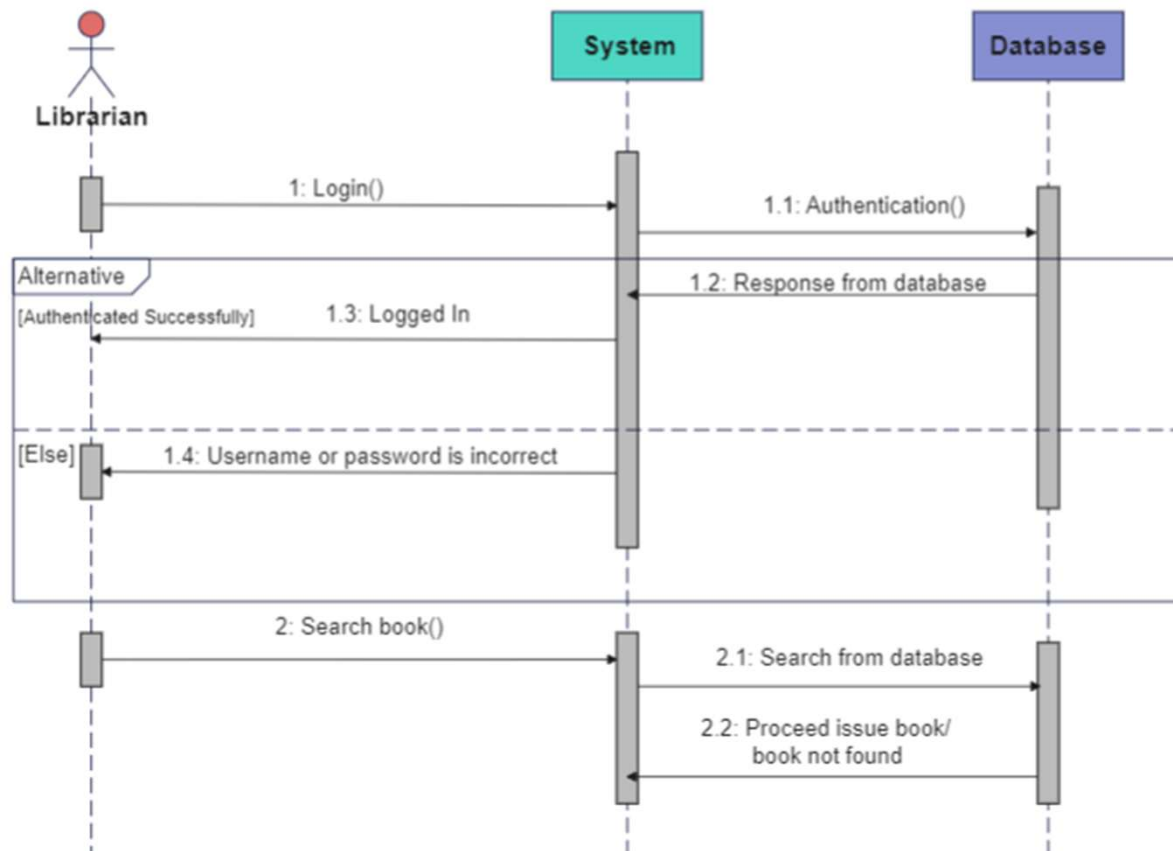
A comment is the ability to attach multiple remarks to an element. It mostly carries useful information for modelers.



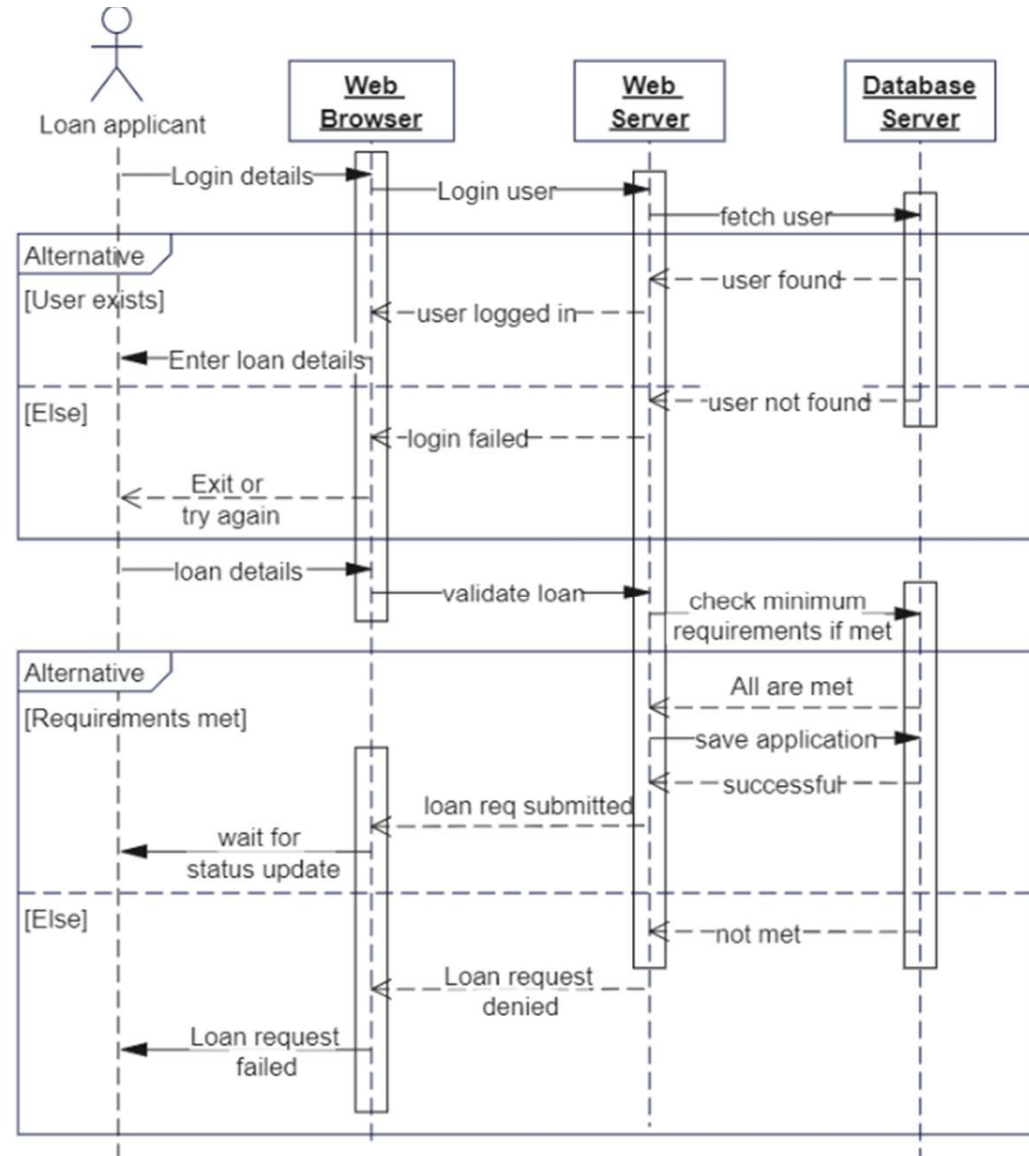
Example 01 - Ticket booking system



Example – 02 – Library Management System



Example – 03 – Loan Application System

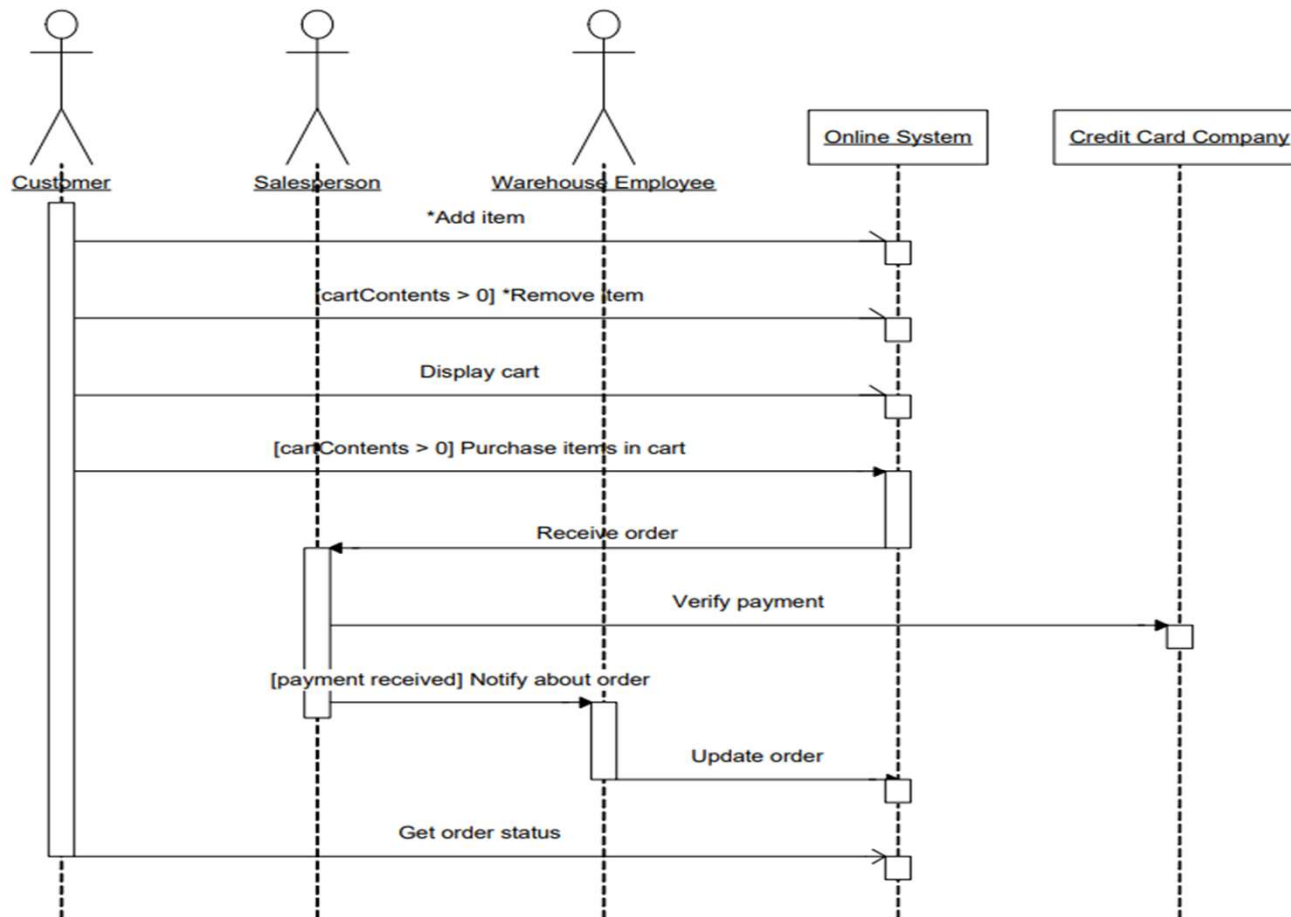


Exercise - 01



Onara Dresspoint sells dresses and jewelries using through online shopping. The customer adds items to her shopping cart. She may remove items or go to the check-out to make her purchases at any time. The customer reviews her purchases, chooses a payment method and pays. A sales employee at Onara Dresspoint gets the order and purchase confirmation from the system, and sends the electronic order to the warehouse. The warehouse employee updates the order status. The customer may check the order status.

Exercise 01 - Solution



Exercise 02 – Online Shopping System



Imagine an online shopping platform where a customer wants to purchase an item. The customer searches for an item, adds it to the cart, and proceeds to checkout. During checkout, the system validates the cart, processes the payment, and confirms the order. Draw a sequence diagram to illustrate the entire process. Include interactions between the customer, search service, shopping cart service, payment service, and order service, detailing how each message is passed and the order in which these interactions occur.

Solution 02 – Online Shopping System

